Lab 8: Cellular Genetics — Mitosis and Meiosis

BIOL-1

Name: ______________________ **Date:** ______________________

Overview

Every cell in your body came from a single fertilized egg that divided over and over again. That process — **cell division** — is how you grow, repair wounds, and replace worn-out cells. But not all cell division is the same:

- **Mitosis** produces identical copies of a cell (for growth and repair).
- **Meiosis** produces unique sex cells called gametes (for reproduction).

In this lab, you will walk through both processes, compare them side by side, discover how meiosis generates the extraordinary genetic diversity that makes every individual unique, and explore what happens when cell division goes wrong.

Learning Objectives

By the end of this lab, you will be able to:

1. Describe the stages of the cell cycle, including interphase and mitosis.
2. Compare what happens in each phase of mitosis vs. meiosis.
3. Explain why meiosis produces genetically unique cells.
4. Identify three sources of genetic variation in sexual reproduction.
5. Describe what happens when cell division goes wrong (cancer, nondisjunction).

Materials

- This worksheet
- Colored pencils (optional, for drawing)

Part 1: Chromosome Basics

Before diving into division, you need the vocabulary. These terms describe what chromosomes look like at different stages of the cell cycle.

1. Match each term to its definition:

Term	Definition
Chromatin	
Chromosome	
Sister Chromatids	
Homologous Chromosomes	
Diploid (2n)	
Haploid (n)	

2. What is the difference between sister chromatids and homologous chromosomes?

Part 2: The Cell Cycle

Before a cell divides, it must prepare. Interphase is the "preparation phase" — the cell spends most of its life here.

3. Fill in what happens during each stage of interphase:

Stage	What Happens
G_1 (Gap 1)	

Stage	What Happens
S (Synthesis)	
G2 (Gap 2)	

4. After S phase, each chromosome consists of two identical copies joined at the centromere. What are these two copies called?

5. What are cell cycle checkpoints? Why are they important?

Part 3: Mitosis — Making Identical Copies

Mitosis is used for growth, repair, and replacing old cells. A skin cell divides by mitosis about every 2–3 weeks.

6. Fill in what happens during each phase of mitosis. Use the memory trick: PMAT.

Phase	What Happens	Memory Cue
Prophase		Chromosomes condense and become visible
Metaphase		Chromosomes line up in the **M**iddle
Anaphase		Chromosomes are pulled **A**part
Telophase		**T**wo new nuclei form

7. What separates during anaphase of mitosis — homologous chromosomes or sister chromatids?

8. After mitosis and cytokinesis, how many daughter cells are produced?

9. Are the daughter cells genetically identical to the parent cell? Are they diploid (2n) or haploid (n)?

10. How does cytokinesis differ between animal cells and plant cells?

Part 4: Meiosis — Making Unique Gametes

Meiosis happens only in the ovaries and testes. It produces egg and sperm cells with half the chromosome number, so that when they fuse at fertilization, the full number is restored.

11. How many rounds of division occur in meiosis? How many daughter cells does meiosis produce from one parent cell?

12. Are the daughter cells diploid or haploid? Are they genetically identical to each other?

13. What separates during Anaphase I?

14. What separates during Anaphase II?

15. Complete the sentence: "Homologs separate in _, *sister chromatids separate in* _."

16. What is crossing over? During which phase does it happen? Why does it matter?

17. What is independent assortment? Why does random orientation of chromosomes at Metaphase I create diversity?

18. Why is it important that gametes are haploid? What would happen if gametes were diploid?

Part 5: Side-by-Side Comparison

19. Fill in the comparison table:

Feature	Mitosis	Meiosis
Number of divisions		
Number of daughter cells		
Ploidy of daughter cells		
Genetically identical to parent?		
Purpose		
Crossing over?		

20. Complete: "Mitosis makes ___ cells that are . *Meiosis makes ___ cells that are* ."

Part 6: Genetic Variation and Real-World Connections

21. Name the three sources of genetic variation in sexual reproduction:

1.
2.
3.

22. In humans (n = 23), how many unique gamete combinations are possible from independent assortment alone? (Calculate 2^{23})

23. What is nondisjunction? Name one condition it can cause (give the chromosome number, e.g., trisomy 21).

24. Using what you learned in this lab, explain in your own words why you are not genetically identical to your siblings (even though you have the same parents).

Part 7: When Cell Division Goes Wrong — Cancer

Cancer is a disease of the cell cycle. When the "brakes" on division fail, cells grow out of control.

25. What are cell cycle checkpoints, and what happens when they fail?

26. What is the difference between a proto-oncogene and an oncogene?

Term	Definition
Proto-oncogene	
Oncogene	

27. What does the tumor suppressor protein p53 normally do? What happens when the p53 gene is mutated?

28. Using the "brakes and accelerator" analogy: in cancer, what role do oncogenes play? What role do mutated tumor suppressors play?